

Subject Description Form

Subject Code	COMP 5517
Subject Title	Human Computer Interaction
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Basic knowledge of programming is required.
Objectives	• To provide students with a broad view of both theoretical and practical issues in human factors for design of human-computer interfaces. • To equip students with knowledge and understanding of the nature of human computer interactions, human characteristics, computer system and interface architecture. • To equip students with sound skills in design, development and evaluation of user interfaces.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. grasp and consolidate the main concepts (conceptual model, metaphors and paradigms) that influence human-computer interaction; b. have a deep understanding of how human cognition and information processing impact human computer interaction; c. critically review and compare different evaluation and testing methods, both quantitative and qualitative; d. apply HCI theory, principles and practices to user interface design to develop creative solutions; and e. systematically analyze an interactive system using appropriate evaluation and usability testing methods.
Subject Synopsis/ Indicative Syllabus	Nature of Human-Computer Interaction (HCI) Definitions and importance of HCI; history and intellectual roots of HCI; roles various disciplines play within HCI. Evaluation Role of evaluation; qualitative and heuristic evaluation techniques; empirical evaluation and benchmarking. Human Characteristics Perception and representation; models and limits of human memory; mental models; use of metaphors; social and organizational aspects; input and output devices: performance characteristics (human and system). Formal and Conceptual Models Task analysis and predictive modeling; dialogue interaction: types and techniques; models for describing interaction processes. Design Guidelines and Metrics

	User-centered design; structural HCI design and envisioning design; standards and metrics; guidelines to support design; standards and metrics; documentation and on-line information. Development and Applications Design rationale; participatory design and prototyping; user interface management systems; WWW application design; groupware; collaborative work and virtual environments. Selected Topics in Advanced HCI Potential topics include: human-robot interaction; ubiquitous computing; speech and natural language interfaces; tangible user interfaces.																																	
Teaching/Learning Methodology	Lectures, Tutorials and Labs The subject material will be delivered through lectures, tutorials and labs. Lectures will provide the main body of the subject materials. Where possible, guest lectures and/or case studies will be used to give the subject material more relevancy to real-world scenarios. Tutorials and labs will provide students with more in-depth opportunities to explore the lecture materials and practice the lecture concepts. Where possible, a hands-on, interactive approach will be used. Projects and Assignments Projects and assignments will provide students with in-depth opportunities to practice the lecture concepts, as well as to assess their ability to apply these concepts in practical scenarios. Examination Examination will assess students on their grasp of subject material.																																	
Assessment Methods in Alignment with Intended Learning Outcomes	Students’ performance in this subject will be assessed by using a letter-grading system in accordance with the University’s convention from grade F (failure) to A+. The relative weights of the different assessment components are as follows: <table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Projects and Assignments</td><td>30</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Examination</td><td>70</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100 %</td><td colspan="5"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	1. Projects and Assignments	30	✓	✓	✓	✓	✓	2. Examination	70	✓	✓	✓	✓	✓	Total	100 %					
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2. Examination	70	✓	✓	✓	✓	✓																												
Total	100 %																																	

	The assignments and examinations will assess students' ability to understand and describe the concepts behind human-computer interaction, as well as various techniques and methods that are used to develop and assess user interfaces. Projects will require students to demonstrate their ability to apply the subject concepts in designing, implementing and analyzing user interfaces.	
Student Study Effort Expected	Class contact:	
	▪ Lectures, tutorials, workshops and labs	39 Hrs.
	Other student study effort:	
	▪ Assignments, Coursework, Reading, Exams	66 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	• A. Dix, J. E. Finlay, G. D. Abowd, and R. Beale, <i>Human-Computer Interaction</i>, 3rd Edition, Harlow: Pearson/Prentice Hall, 2004. • D. Norman, <i>The Design of Everyday Things</i>, Revised and Expanded Edition, New York: Basic Books, 2013. • B. Shneiderman, C. Plaisant, M. Cohen, S. Jacobs, N. Elmqvist, and N. Diakopoulos, <i>Designing the User Interface: Strategies for Effective Human-Computer Interaction</i>, 6th Edition, Boston: Pearson, 2017. • W. J. Smith, <i>ISO and ANSI Ergonomic Standards for Computer Products: A Guide to Implementation and Compliance</i>, Upper Saddle River, N.J.: Prentice Hall, 1996. • ACM Transactions on Graphics • IEEE Transactions on Visualization and Computer Graphics • ACM Transactions on Computer-Human Interaction • ACM Conference on Human Factors in Computing Systems • ACM Symposium on User Interface Software and Technology • ACM Symposium on Interactive 3D Graphics • ACM Symposium on Virtual Reality Software and Technology • IEEE International Symposium on Mixed and Augmented Reality • ACM SIGGRAPH • ACM SIGGRAPH VRCAI • IEEE Virtual Reality • IEEE VIS • IEEE Transactions on Human-Machine Systems • International Journal of Human Computer Studies • International Journal of Human-Computer Interaction • International Journal of Child-Computer Interaction • Human Computer Interaction • Interacting with Computers • Computer Graphics Forum • Computers & Graphics • Graphical Models • Computer Animation and Virtual Worlds	